

Fig. 4: Circuit for regulated power supply

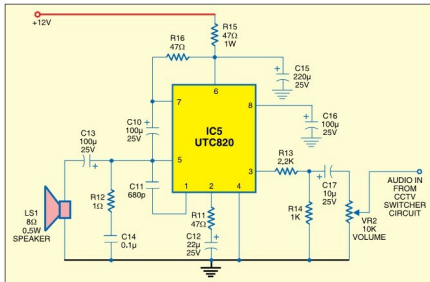


Fig. 5: Circuit diagram of power amplifier

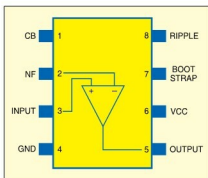


Fig. 6: Pin diagram of UTC820 audio power amplifier

for the built-in camera module. The 12V power supply is permanently connected to all the cameras. Audio and video signals from the cameras are selected using switch pairs S2a-S2b through S7a-S7b (built inside IC2

through IC4) and fed to the audio power amplifier and video output sockets, respectively.

Each IC has four CMOS bilateral switches for transmission or multiplexing of analogue or digital signals. It is pin-to-pin compatible with IC CD4016, but has much lower 'on' resistance, which is relatively constant over the input signal range.

Video and audio signals from SK1 (camera 1) are connected to S2a and S2b, respectively, and both the enable pins (to switch on both the switches) are shorted out and connected to port RC7 (pin 18) of the microcontroller. Similarly, audio and video signals from SK2 through SK6 are connected to IC2 through IC4 and then to Port C of IC1 as shown in Table I.

Enable pins of switch pairs S2a-S2b, S3a-S3b, S4a-S4b, S5a-S5b, S6a-S6b and S7a-S7b of IC2 through IC4 are shorted to each other and fed to the corresponding port of Port C as shown in Table I. The microcontroller provides high logic at Port-C pins to switch on a particular camera. The truth table for Port C is shown in Table II.

Immediately after switching on the mains, the standby LED (LED7) connected to Port-C pin RC3 glows and the LCD shows the welcome page (refer to the LCD screenshot).

On pressing the power key on the remote, the standby LED turns off and the LCD shows which camera is 'on,' in which mode it is now and default hold time for the corresponding camera in 'Auto' mode.

As described above, pins of Port C (RC0, RC1, RC4-RC7) are connected to IC2 through IC4 for selecting the camera to be switched on. LED1 through LED6 are used to check the logic status of Port C. Pressing keys '1' through '6' on the remote makes the corresponding LEDs (LED1 through LED6) glow, indicating 'on' condition of the cameras (CAM-1 through CAM-6).

Here a 16-character, 2-line LCD is used in 4-bit mode. Data lines D4 through D7 of the LCD are connected to Port-B pins RB4 through RB7. Enable (EN) and reset (RS) pins are connected to RB2 and RB3 pins, respectively, and the read/write (R/W) pin is connected to ground as we have used the LCD in writing mode only. Preset VR1 (10-kilo-ohm) is used for setting the LCD character contrast.

Power supply

A simple circuit for regulated supply (shown in Fig. 4) is used here. The +12V regulated output is used to power all the six cameras, IC2 through IC4 and IC5 and fed to another 7805 voltage regulator (IC7) to get the regulated +5V output for the microcontroller (IC1) and the LCD.

The current rating of the transformer should be around 2A, depending on the current consumption of the